

# Section 11 Solution

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## 4 Exercises

### 4.1 Fun with cost minimization

For each of the following functions, derive the total cost  $c(w, r, y)$ , average cost  $\frac{c(w, r, y)}{y}$ , and marginal costs  $\frac{\partial c(w, r, y)}{\partial y}$ . We'll go over these more next time.

1.  $f(L, K) = 2L^{\frac{1}{4}}K^{\frac{1}{4}}$

**Answer:**

$$\begin{aligned}\mathcal{L}(w, r, y) &= wL + rK - \lambda(f(L, K) - y) \\ 0 &= \frac{\partial \mathcal{L}}{\partial L} = w - \lambda \frac{1}{2} L^{-\frac{3}{4}} K^{\frac{1}{4}} \\ 0 &= \frac{\partial \mathcal{L}}{\partial K} = r - \lambda \frac{1}{2} L^{\frac{1}{4}} K^{-\frac{3}{4}} \\ 0 &= \frac{\partial \mathcal{L}}{\partial \lambda} = y - 2L^{\frac{1}{4}}K^{\frac{1}{4}}\end{aligned}$$

$$\begin{aligned}y &= 2L^{\frac{1}{4}} \left(\frac{r}{w}\right)^{\frac{1}{4}} L^{\frac{1}{4}} \\ L &= \frac{1}{4} \left(\frac{r}{w}\right)^{\frac{1}{2}} y^2 \\ K &= \frac{1}{4} \left(\frac{w}{r}\right)^{\frac{1}{2}} y^2\end{aligned}$$

$$\begin{aligned}c(w, r, y) &= \frac{1}{2} y^2 (rw)^{\frac{1}{2}} \\ \frac{c(w, r, y)}{y} &= \frac{1}{2} y (rw)^{\frac{1}{2}} \\ \frac{\partial c}{\partial y} &= y (rw)^{\frac{1}{2}}\end{aligned}$$

2.  $f(L, K) = [\min(L, K)]^{\frac{1}{3}}$

**Answer:** Using a Lagrangian as a tool to solve this exercise is not going to be helpful. Why? FOC help us to identify an interior solution of the problem. In the cost minimization problem is equivalent to the point in which the ratio of prices of the inputs is equivalent to the MRTS (or ratio of marginal productivities): a point in which we have tangency between an isoquant and the border of the expenditure set.

Going back to the  $[\min(L, K)]^{\frac{1}{3}}$  production function, we saw last section that in the corner of its isoquants the MRTS is not defined.

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We will need to take an ad-hoc approach. For this function, we can see that to minimize cost it must be that  $L^* = K^* = y^3$ . Therefore the cost function is

$$\begin{aligned} c(w, r, y) &= (w + r) y^3 \\ \frac{c}{y} &= (w + r) y^2 \\ \frac{\partial c}{\partial y} &= 3(w + r) y^2 \end{aligned}$$

## 4.2 Problem 2, MT2 Fall 2004

In this exercise, we consider a firm producing product  $y$  using two inputs, labor  $L$  and capital  $K$ . The production function is  $y = f(L, K) = (L + K)^\alpha$ . Assume that the wage of a worker is  $w$  and the cost of capital is  $r$ . Assume  $L \geq 0$ ,  $K \geq 0$ , and  $\alpha > 0$ .

1. Draw a picture of the isoquants. What is the unusual feature of this production function?

**Answer:**

In this particular firm, the inputs are perfect substitutes. The firm can perfectly substitute one unit of capital for one unit of labor and viceversa.

2. For which values of  $\alpha$  does the function exhibit decreasing returns to scale (that is,  $f(tL) < tf(L)$  for all  $t > 1$  and all  $L \geq 0$ )?

**Answer:**

For this production function,  $f(tL, tK) = (tL + tK)^\alpha = t^\alpha (L + K)^\alpha = t^\alpha f(L, K)$ . Therefore, for  $0 < \alpha < 1$  the firm will exhibit decreasing returns to scale since  $t^\alpha < t$  for  $t > 1$  and  $\alpha < 1$ .

3. Consider now the first step of the cost minimization problem. The firm solves

$$\min wL + rK \quad \text{s.t. } f(L, K) \geq y$$

for  $y > 0$ . What are the solutions for  $L^*(w, r, y|\alpha)$  and  $K^*(w, r, y|\alpha)$ ? (This notation stresses that the solution depends also on the parameter  $\alpha$ . Hint: You are better off not using Lagrangeans. The pictures you drew in point 1 may be helpful)

**Answer:**

From Point 1, we know that the firm is indifferent between a unit of labor and a unit of capital. Therefore, in its cost minimization problem the firm will just choose the input that is cheaper. That is, the firm will use only  $L$  if  $w < r$  and only  $K$  if  $w > r$ . For  $w = r$ , the firm is indifferent between any combination of labor and capital. One can see this graphically from the Figure in point 1. The isoquants are straight lines, so like in the case of expenditure minimization, the solution will be in the corners. To fully solve the problem, consider first the case  $w < r$ . In this case the firm uses only labor and the cost minimization becomes

$$\min wL \quad \text{s.t. } L^\alpha \geq y$$

We already solved this in class. We know that the constraint will be binding and the firm will use just about enough labor to produce  $y$ . Therefore,  $L^*(w, r, y|\alpha)$  is the solution of  $L^*(w, r, y|\alpha)^\alpha = y$ , or  $L^*(w, r, y|\alpha) = y^{1/\alpha}$  and  $K^*(w, r, y|\alpha) = 0$ . Consider now the case  $w > r$ . Now the cost minimization becomes

$$\min rK \quad \text{s.t. } K^\alpha \geq y$$

Again, the firm will use just about enough labor to produce  $y$ . Therefore,  $K^*(w, r, y|\alpha)$  is the solution of  $K^*(w, r, y|\alpha)^\alpha = y$ , or  $K^*(w, r, y|\alpha) = y^{1/\alpha}$  and  $L^*(w, r, y|\alpha) = 0$ . Finally, for  $r = w$ , there are multiple solutions for  $L^*(w, r, y|\alpha)$  and  $K^*(w, r, y|\alpha)$ , all satisfying  $L^*(w, r, y|\alpha) + K^*(w, r, y|\alpha) = y^{1/\alpha}$ .

4. Write down the implied cost function  $c(w, y|\alpha)$ .

**Answer:**

The cost function  $c(w, y|\alpha)$  is  $wL^*(w, r, y|\alpha) + rK^*(w, r, y|\alpha)$ , which equals

$$c(w, y|\alpha) = \begin{cases} ry^{1/\alpha} & \text{if } w > r \\ wy^{1/\alpha} = ry^{1/\alpha} & \text{if } w = r \\ wy^{1/\alpha} & \text{if } w < r \end{cases}$$

or, in a more compact format,  $c(w, y|\alpha) = \min(w, r) * y^{1/\alpha}$ .

5. Derive an expression for the average cost  $c(w, y|\bar{L}, \alpha)/y$  and the marginal cost  $c'_y(w, y|\bar{L}, \alpha)$  for  $y > 0$  and  $w < r$ . Graph the average cost and marginal cost for  $\alpha = .5$ ,  $w = 1$ , and  $r = 2$ . Graph the supply function for the same values of the parameters. [remember,  $y$  is on the horizontal axis].

**Answer:**

Under the assumption  $w < r$ , the cost function is  $wy^{1/\alpha}$ . The average cost function  $c(w, y|\bar{L}, \alpha)/y$  is  $wy^{1/\alpha}/y = wy^{(1-\alpha)/\alpha}$ . The marginal cost function  $c'_y(w, y|\bar{L}, \alpha)$  is  $w\frac{1}{\alpha}y^{(1-\alpha)/\alpha}$ . For  $\alpha = 1/2$ , the average cost function is  $wy$  and the marginal cost function is  $2wy$ . Both are increasing, and the marginal cost function lies always above the average cost function. This implies that the supply function is just the marginal cost function.

6. Now that we graphically solved for the supply function, we also derive it formally for all  $\alpha > 0$ . Consider the second step of cost minimization

$$\max_y py - c(w, y|\bar{L}, \alpha).$$

Write down the first order condition and the second order conditions. Solve for  $y^*(w, p|\bar{L}, \alpha)$ . (here do *not* assume  $w < r$ ) For what values of  $\alpha$  is the second order condition satisfied?

**Answer:**

The first order condition is

$$p - c'_y(w, y|\bar{L}, \alpha) = 0$$

or

$$p - \frac{1}{\alpha} \min(w, r) y^{(1-\alpha)/\alpha} = 0.$$

This implies

$$y^*(w, y|\bar{L}, \alpha) = p^{\frac{\alpha}{1-\alpha}} \left( \frac{\alpha}{\min(w, r)} \right)^{\frac{\alpha}{1-\alpha}} \quad (1)$$

The second order condition is

$$-c''_{y,y}(w, y|\bar{L}, \alpha) < 0$$

or

$$-\min(w, r) \frac{(1-\alpha)}{\alpha^2} y^{(1-2\alpha)/\alpha} < 0.$$

This condition is satisfied for  $\alpha < 1$ .

7. From now on, assume  $\alpha < 1$ . Take the solution for  $y^*(w, p|\bar{L}, \alpha)$  in point 6 and consider what happens to  $y^*$  as the wage  $w$  increases. Obtain the sign of  $\partial y^*/\partial w$  for the cases  $w < r$  and  $w > r$ . Provide intuition on this result.

**Answer:**

Expression (1) for  $y^*$  implies  $\partial y^*/\partial w < 0$  for  $w < r$ , since in this case  $y^* = p^{\frac{\alpha}{1-\alpha}} \left( \frac{\alpha}{w} \right)^{\frac{\alpha}{1-\alpha}}$ . However,  $\partial y^*/\partial w = 0$  for  $w > r$ , since in this case  $y^* = p^{\frac{\alpha}{1-\alpha}} \left( \frac{\alpha}{r} \right)^{\frac{\alpha}{1-\alpha}}$ , and  $w$  does not show up in this expression. Increases in the cost of labor only matter in the supply function as long as labor is used, that is, as long as it is the cheapest input ( $w < r$ ). If labor is not the cheapest input, instead, small increases in labor costs do not matter since no labor is employed.

8. Consider now what happens to the supply function as price of output  $p$  increase. Obtain the sign of  $\partial y^*/\partial p$  and provide intuition on the result.

**Answer:**

Expression (1) for  $y^*$  implies  $\partial y^*/\partial p = \frac{\alpha}{1-\alpha} p^{\frac{\alpha}{1-\alpha}-1} \left( \frac{\alpha}{\min(w,r)} \right)^{\frac{\alpha}{1-\alpha}} > 0$ . As the price of the final good increases, the company finds it profitable to supply more output to the market. The higher final price makes it worthwhile to produce even if increasing production raises the marginal cost.

9. Does the company make, negative, or positive profits for  $p > 0$ ? Provide an argument for your answer.

**Answer:**

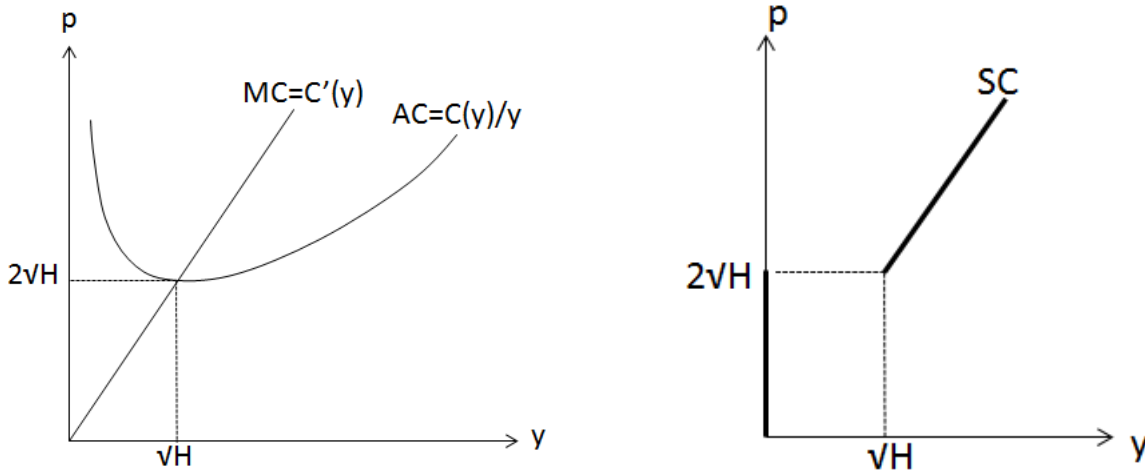
Given that the marginal cost function always lies above the average cost function (see point 5) the firm is making positive profits for all levels of  $p$ , as long as  $p > 0$ .

### 4.3 Production with Fixed Costs

Suppose a firm has a cost function  $c(y) = H + y^2$  if  $y > 0$  and  $c(0) = 0$ .

1. Derive and draw the  $MC$  and  $AC$  curves.

**Answer:**  $MC = C'(y) = 2y$ ,  $AC = C(y)/y = H/y + y$ . See graph below on the left.



2. Solve for  $y$  at the point where the two curves intersect, and show that this point is also the minimum of the  $AC$  curve.

**Answer:** In fact, this must always be so. Think about the intuition. We can prove it using the quotient rule to differentiate  $\frac{c(y)}{y}$  with respect to  $y$  (this is the f.o.c. for finding the minimum of  $\frac{c(y)}{y}$ ):

$$\frac{\partial}{\partial y} \left( \frac{C(y)}{y} \right) = \frac{C'(y_{min})y_{min} - C(y_{min})}{y_{min}^2} = 0$$

$$C'(y_{min}) = \frac{C(y_{min})}{y_{min}}$$

3. Draw the supply function for the firm with cost function  $c(y) = H + y^2$  if  $y > 0$  and  $c(0) = 0$ , and write an expression representing it:  $(y^*(p))$ .

**Answer:**

See graph above on the right.

$$y^*(p) = \begin{cases} p/2 & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases}$$

4. If  $p$  stays constant, how does  $y^*$  respond to an increase in  $H$ ?

**Answer:**

Say that  $H$  increases to  $H'$ . There are three possible scenarios:

- $2\sqrt{H'} \leq p$ , then  $y^*$  stays at  $p/2$ .
- $2\sqrt{H} \leq p < 2\sqrt{H'}$ , then  $y^*$  falls from  $p/2$  to 0.
- $p < 2\sqrt{H}$ , then  $y^*$  stays at 0.

5. What is the profit function of this firm?

**Answer:**

$$\pi^*(p) = \begin{cases} py - H - y^2 & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases} = \begin{cases} p^2/4 - H & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases}$$